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NBA Betting: Are the Best Performing NBA Teams Actually Bad Bets?

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ABSTRACT

As legalized sports betting continues to expand across America, understanding how betting markets interpret team performance has become increasingly important. In the National Basketball Association (NBA), traditional metrics such as win percentage and scoring efficiency are widely used to evaluate team strength, yet their effectiveness in predicting betting success against the point spread remains unclear. This study explores whether observable team performance translates into an advantage over sportsbook expectations.

Leveraging historical NBA game data and betting market information from 2007–2025, this research develops a structured framework for evaluating team performance relative to the spread. By integrating performance metrics, betting context variables, and classification modeling techniques, the study shifts the focus from predicting game winners to assessing whether teams exceed market expectations. This work contributes to sports analytics by examining the relationship between team strength and betting efficiency, offering a new perspective on how performance is priced into modern sports betting markets.

INTRODUCTION

The expansion of legalized sports betting across the United States has significantly increased interest in understanding market behavior, particularly in professional sports leagues such as the NBA. As betting has become easily accessible, both casual and data-driven bettors seek strategies to gain an edge over sportsbooks. A central challenge in this space is determining whether publicly observable team performance translates into profitable betting opportunities.

In the NBA, team strength and performance are commonly measured using metrics such as win percentage, point differential, and offensive and defensive efficiency. While these indicators are highly predictive of game outcomes, sportsbooks incorporate this same information when setting betting lines. As a result, the primary question is not simply which teams win, but whether they exceed expectations relative to the betting market.

Research Question

Are high-performing NBA teams profitable bets when considering betting spreads and odds? This study evaluates whether strong teams consistently cover the point spread rather than simply winning games over the 2007–2025 seasons.

Mechanics of the Spread

A point spread is a handicap set by sportsbooks to balance betting between two teams. The favored team must win by more than the spread to 'cover', while the underdog covers by winning outright or losing by fewer points than the spread. This mechanism reflects expected performance differences and aims to create an even betting market.

Importance to the NBA and Casual Bettors

Betting markets serve as a real-time reflection of team performance, public perception, and information efficiency. Efficient markets suggest that all relevant information—including team strength, injuries, and

game context—is already incorporated into the spread. For casual bettors, misunderstanding the role of the spread is a frequent mistake. Many assume that strong teams are inherently good bets because they win frequently; however, because sportsbooks adjust spreads to reflect team strength, consistently betting on heavy favorites may not yield long-term profitability.

This study evaluates whether betting markets fully account for team strength or whether exploitable inefficiencies remain, especially in games involving large spreads.

DATA AND METHODS

The dataset was constructed by joining two public historical archives covering NBA regular and postseason games from October 2007 through 2025:

1. Historical NBA Box Scores: Game-level box scores and player performance statistics.

2. Historical NBA Betting Data: Closing spreads, over/under totals, and favorite/underdog designations across sportsbook operators.

Variable Definitions and Feature Engineering

Table 1 summarizes the primary attributes and engineered metrics utilized in the modeling workflow.

Variable Name	Type	Description
gameId	Identifier	Unique identifier for each specific NBA game.
gameDateTimeEst	Timestamp	Date and time the game was played (EST).
full_team_name	Categorical	Primary team being tracked in the record.
full_opponent_team_name	Categorical	Opposing team name.
home	Binary	Indicator: 1 for Home team, 0 for Away team.
win	Binary	Indicator: 1 for a Win, 0 for a Loss.
spread	Continuous	Point handicap set by oddsmakers.
whos_favored	Categorical	Team expected to win by the betting market.
overall_offensive_score	Continuous	Total points scored: (3PM × 3) + (2PM × 2) + FTM.
overall_defensive_score	Continuous	Defensive metric: DefRebound + Block.
overall_team_peformance_score	Continuous	Aggregated composite performance rating.
team_level_of_performance	Categorical	Ranked tier (High, Medium, Low) based on 33rd and 66th percentiles.
cover_spread	Binary Target	1 if Margin of Victory > Point Spread; 0 otherwise.

Table 1. Description of Key Dataset Variables and Engineered Features

Team Performance Composite Metric

To measure overall game contribution, an aggregated performance index was calculated using Points Per Possession (PPP ≈ 1.10):

PPP = Sum(teamScore) / Sum(fieldGoalsAttempted + 0.44 × freeThrowsAttempted)

The constant 0.44 accounts for possessions where free throws do not consume a complete offensive trip. Using this factor, performance components were weighted:

• Performance = TotalScore + (2.2 × Defensive Rebounds) + (1.1 × Blocks)

• Games were classified into Low Tier (< 33rd percentile), Medium Tier (33rd–66th percentile), and High Tier (> 66th percentile) based on this index.

Target Variable Formulation

The target variable is binary spread coverage:

• Cover_spread = 1 if (Points Scored - Points Allowed) > Sportsbook Spread

• Cover_spread = 0 otherwise

STATISTICAL ANALYSIS AND MODELING

Historical Disruptions and Home-Court Context

Two major structural disruptions occurred during the study period: the 2011 NBA labor lockout (shortened season) and the 2020 COVID-19 pandemic hiatus and bubble format. Both caused pronounced temporary dips in cumulative volume metrics such as total field goals made.

Additionally, while home teams achieved a higher outright win percentage (approximately 58% vs. 42% away), the spread coverage rate showed minimal variance between home favorites, away favorites, home underdogs, and away underdogs (fluctuating tightly within the 48%–50% range). Sportsbooks efficiently factor geographic home advantage into spread pricing.

Empirical Coverage Rates Across Performance Tiers

Evaluating cover percentages across the 2007–2025 dataset revealed a sharp gradient based on team performance tiers:

• **High-Performing Teams:** Covered the spread 88.47% of the time.

• **Medium-Performing Teams:** Covered the spread 66.07% of the time.

• **Low-Performing Teams:** Covered the spread only 36.66% of the time.

Starting around 2015—aligning with the widespread integration of advanced analytics across betting markets—cover rates for lower-tier teams declined even further (dropping from 40%–45% down to 13%–20%), while top-tier teams consistently maintained cover rates between 83% and 95%.

Predictive Modeling

Two supervised classification models were trained on a 70/30 train-to-test split. Missing numeric values were imputed using column means, and categorical features were one-hot encoded.

Logistic Regression: Logistic regression was estimated to evaluate continuous probabilities of covering the spread. The model achieved an Accuracy of 0.76 and an AUC-ROC of 0.8235. Shooting efficiencies (three-point percentage, two-point percentage, and free-throw percentage) and composite performance scores exhibited positive coefficients. Notably, the interaction between team performance and spread size was positive: as team quality increased, the typical penalty of covering a wider handicap was substantially mitigated.

Decision Tree Classifier: A decision tree was trained using grid-search hyperparameter tuning, identifying an optimal maximum depth of 6 to prevent overfitting on game-level noise. While the

unconstrained baseline tree achieved Accuracy of 0.70 and AUC-ROC of 0.6704, the constrained tree (max_depth = 6) attained Accuracy of 0.74, AUC-ROC of 0.81, Precision of 0.78, and Recall of 0.85.

Feature importance evaluation confirmed that overall_team_peformance_score accounted for the vast majority of tree splits (>0.70 importance weight), followed by spread and threePointersPercentage.

Model Specification	Accuracy	AUC-ROC	Precision (Cover)	Recall (Cover)
Logistic Regression	0.76	0.824	—	—
Decision Tree (Default)	0.70	0.670	0.77	0.76
Decision Tree (max_depth = 6)	0.74	0.810	0.78	0.85

Table 2. Model Performance Summary Comparison

CONCLUSION

This study investigated whether high-performing NBA teams represent profitable bets when factoring in sportsbook point spreads. Rather than demonstrating that sportsbooks overprice elite teams, the empirical evidence demonstrates the opposite: high-performing teams consistently beat market expectations, covering the spread at an 88.47% rate over nearly two decades.

Key conclusions include:

1. Spread Inefficiencies at Extremes: Sportsbooks adjust lines for elite teams, but they often do not adjust them enough to neutralize superior offensive and defensive efficiency.

2. Impact of Shooting Efficiency: Three-point, two-point, and free-throw efficiencies serve as primary drivers of covering margins.

3. Resilience to Large Spreads: The positive interaction between team performance and spread size indicates that heavy favorites remain strong betting propositions.

Future Research and Next Steps

• **Operational:** Constructing a real-time tracking pipeline to compute live in-game performance metrics and assess dynamic spread opportunities.

• **Research Extensions:** Incorporating travel schedules, rest differentials, and back-to-back scheduling fatigue to assess whether cover drop-offs disproportionately affect specific performance tiers.

REFERENCES

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